

AVO Valve Characteristic Meter MK III

Calibration manual

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Important!

This document is an unofficial working aid for a specific AVO VCM Mk III procedure. It does not replace the original manufacturer's documentation, safe working practices or verification against the actual wiring in your instrument. This procedure involves hazardous voltages. If you choose to proceed, you do so entirely at your own risk.

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Field	Value
Instrument	AVO Valve Characteristic Meter MK III
Serial number	
Owner/workshop	
Calibrated by	
Measurement equipment	
Mains voltage during calibration	

Revision History

Version	Date	Change
v1.0	2026-07-04	First version

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1. Safety and scope

Purpose

The aim is to make the calibration repeatable, documentable and easier to follow on the bench.

Basic principle

- Measure first, adjust last. Many steps are checks and must not be hidden by trimming the wrong adjustment.
- If a check point is wrong: work backwards through the meter, SET ~, critical resistors, switches and previous modifications.
- Record original values before replacing components or making adjustments.
- Keep the original circuit separate from later modifications, especially meter protection, rectifier diodes and C1.

High voltage

The AVO VCM MK III contains mains voltage and high secondary voltages. Prefer connecting probes with the instrument switched off, use insulated clips, keep one hand away from the instrument during live measurements and discharge capacitors before cold measurements.

2. Tools and components

Basic equipment

Equipment	Minimum requirement / recommendation	Used in steps
DMM with DC μA range	Must measure around 30 μA DC with sufficient resolution, preferably 0.1 μA or better. Check fuse and the correct $\mu\text{A}/\text{mA}$ input. Examples: Keysight U1242B, bench DMM or any comparable DMM with a μA range.	1
General high-impedance DMM	DC V, DC mV, AC RMS, resistance and continuity. Input impedance at least 2 M Ω for grid-voltage checks; 10 M Ω preferred. Do not use LoZ mode for NEG. GRID VOLTS or mA/V voltage checks	2, 4, 6, 7, 8, 12
External DC mA meter	Must read 50 mA DC stably in the A1 ANODE LINK, preferably with 0.1 mA resolution or better and a fused current input. May be a separate DMM or bench meter.	5
Adjustable 0-30 V DC supply	For standalone meter movement test through 470 k. Current limiting is useful. Always start at 0 V.	1
Insulated test leads and crocodile clips	Insulation rating suitable for the voltages being measured. Prefer connecting with the instrument switched off.	All
Insulated trimming tool	For RV4/RV6 if internal adjustment is required. Use only with a safe method.	3, 6
Record sheet/notebook	Record measurements, replaced components and adjustments.	All

Note on measurement equipment

No specific meter model is required. What matters is the correct range, resolution, input impedance, fuse protection and safe connection. For the grid voltage and mA/V voltage checks, use a high impedance electronic voltmeter or DMM. The grid voltage source is relatively high impedance, so the voltmeter input resistance should be at least 2 M Ω ; a normal 10 M Ω DMM is preferred. Do not use LoZ mode, and avoid analogue 20,000 Ω/V meters such as an AVO Model 8 for the grid voltage calibration, as they may load the circuit and give a falsely low reading. A DMM without a DC μA range cannot perform step 1 as a direct current measurement; it can only be used indirectly by measuring voltage across a known precision resistor.

Test components and fixtures

Part	Specification	Use
470 k safety resistor	0.25 W or higher	Step 1, in series with the DC supply and meter.
10 k reference resistor	0.1 % recommended, 1 % acceptable	Alternative indirect current measurement and check.
Dummy-load fixed resistor	3.9 k, at least 10 W	Step 5, between A1 and C in series with diode and pot.
Dummy-load potentiometer	1 k, approx. 3 W, wirewound	Step 5, adjusts load to 50 mA DC.
Dummy-load diode	1 kV silicon diode, e.g. 1N4007	Step 5. Diode anode to A1, band/cathode towards resistor/C side.
Temporary mA/V resistor	220 k, 0.25 W or higher	Step 7, temporarily in series with the live end of RV5/grid pot feed.
D/R test resistor	150 ohm, at least 5 W	Step 10, in series with test diode.
D/R test diode	1N4007 or equivalent 1 kV diode	Step 10.
Octal test plug	With 1 % high-value resistors	Step 8, leakage/resistance scale.

Octal test plug for leakage scale

Octal pin	Resistor to pin 1	Comment
1	Common / 0 ohm	Common point.
2	0.5 Mohm	Low part of leakage scale.
3	1 Mohm	Classic AVO check value.
4	2.5 Mohm	Practical mid-scale value.
5	5 Mohm	Scale check.
6	10 Mohm	Scale check.
7	25 Mohm	High part of scale.
8	NC	Not used

Permanent components commonly checked

Part/function	Original / target	Comment
R35 meter shunt	10 k	Do not duplicate if a meter protection PCB already includes a shunt.
Meter + swamp	30 uA FSD, 3250 ohm +/- 1 %	Base condition for all later calibration.
R19A/R19B	Total 2.96 Mohm	Must not be changed just to make SET ~ appear correct.
R7-R14	2.5 ksteps	Affects NEG. GRID VOLTS coarse steps.
R15	3.54 k in parallel with RV5	Affects the low and mid part of the grid scale.
RV5 NEG. GRID VOLTS	10 k linear wirewound	Mechanical indexing is important.
RV3 SET mA/V	10 k linear wirewound	Check especially if mA/V calibration is off.
RV1/RV2 backing off	25 ohm / 250 ohm	Clean and check before replacement.
C1	8 uF original, high-voltage	Replacement must be deliberate, often 8.2 uF film or 8-10 uF electrolytic >=450 V.
D77/6AL5	V1/V2	Check valve function and socket contact.

3. Component location quick guide

Appendix A shows the circuit diagram and Appendix B shows the MK III layout. Always follow the actual wiring of your instrument, since previous repairs and modifications may differ.

Note on RV3/RV5:

In the original Control Panel parts list, the Circuit Ref column identifies Set mA/V as RV5 and Negative Grid Volts as RV3. This appears to be reversed compared with the circuit diagram and calibration procedure. In this revised table, the circuit references have therefore been corrected to match the schematic/function rather than the original parts-list entry.

Function / part	Where to start looking	Comment
Meter movement and R35	Circuit diagram near the meter; layout Plate 3 Control Panel	R35 is the 10 k shunt in the meter circuit.
C1	Layout Plate 2 Power Supply Panel	High-voltage capacitor in the circuit. Discharge and measure safely.
RV4 Cal. Grid Volts	Layout Plate 2 Power Supply Panel; circuit diagram near V1/D77 and grid link	Internal preset for step 6. Dangerous location in MkIII.
RV6 Cal. Set ~	Layout Plate 2 Power Supply Panel; circuit diagram near SET ~ section	Only adjusted if SET ~ is not correct after previous checks.
RV5 NEG. GRID VOLTS	Front/control panel Negative Grid Volts control	10 k linear. Check mechanical stop and scale.
RV3 SET mA/V	Front/control panel SET mA/V control	10 k linear. Used in step 7.
RV1/RV2 Backing Off	Control Panel	25 ohm and 250 ohm wirewound.
V1/V2 D77/6AL5	Power Supply Panel and screen relay area	Used in grid/screen functions, not as a normal smoothed anode rectifier.
A1/A2/S/G/C	Top/front terminals and valve holder panel	Use socket and pin diagrams for real valve tests.

4. Calibration steps 1-12

Step 1 - Meter movement check

Purpose

Verify the panel meter mechanical zero, full-scale deflection and resistance before any other calibration. If the meter is not correct, all later adjustments will be misleading.

Setup / settings

- AVO switched off and disconnected from mains.
- Meter left in its normal sloping panel position.
- One side of the meter disconnected from the VCM circuit so the movement can be tested in isolation.
- Adjustable 0-30 V DC supply, 470 k in series and a DMM with DC uA range.
- Optional: separate DMM on DC mV directly across the meter at FSD.

Procedure

1. Set the mechanical zero with no current using the zero adjust until the pointer rests at 0 on the 0-100 scale.
2. Connect: DC+ -> 470 k -> DMM on DC uA -> AVO meter -> DC-.
3. Start the DC supply at 0V and increase slowly.
4. Check roughly 7.5 uA at 25, 15 uA at 50 and 22.5 uA at 75 on the scale.
5. Stop when the pointer is exactly at 100 on the main scale. Do not push against the mechanical stop.
6. Read the uA meter. Target is 30.0 uA.
7. Preferably measure the voltage across the meter. About 97.5 mV at 30 uA corresponds to 3250 ohm.
8. Reduce the DC supply to 0V and check that the pointer returns smoothly to zero.

Target values / assessment

Point	Target value / comment
Mechanical zero	0 on the 0-100 scale with no current.
FSD	30.0 uA at 100 on the main scale.
Meter + swamp	3250 ohm +/-1 %, indirectly by V/I.
Voltage across meter at FSD	approx. 97.5 mV.
If R35 remains in parallel	Total current will be about 39.75 uA; isolate the movement for the 30 uA test.

Notes

- Mechanical zero adjust is for zero only, not to make full scale correct.
- 470 k current-limits the test; about 14-16 V from the DC supply should be enough.

Record

Point	Target	Measured	Comment
Mechanical zero	0		
Full scale	30.0 uA		
Voltage across meter	approx. 97.5 mV		
Calculated resistance	3250 ohm +/-1 %		

Step 2 - Cold component check

Purpose

Ensure that calibration does not mask faulty resistors, potentiometers, capacitors, switches or previous modifications.

Setup / settings

- AVO fully switched off and discharged.
- DMM on resistance and continuity.
- Photograph before disconnecting components or wires.
- Lift one end of components that cannot be measured correctly in -circuit.

Procedure

1. Measure R35, 10 k meter shunt. Check that it is not duplicated by a meter protection PCB.
2. Measure R19A/R19B or equivalent calibration resistors, total 2.96 Mohm.
3. Measure R7-R14, the 2.5 k steps for NEG. GRID VOLTS.
4. Measure R15, 3.54 k in parallel with RV5/grid pot.
5. Measure RV5 NEG. GRID VOLTS and RV3 SET mA/V through their full travel.
6. Check RV1/RV2 backing off for open spots, jumps and mechanical wear.
7. Check RV4 and RV6 presets for open circuit/burnt track.
8. Check C1: value, polarity if electrolytic, leakage and mechanical condition.
9. Check D77/6AL5 and any diode/rectifier modifications against the schematic.
10. Check fuse, mains selector, loose wires and cracked solder joints.

Target values / assessment

Point	Target value / comment
R35	10 k.
R19 total	2.96 Mohm.
R7-R14	2.5 k each.
R15	3.54 k.
RV5/RV3	10 k linear, smooth wiper contact.
RV1/RV2	25 ohm / 250 ohm backing off.
RV4/RV6	5 k presets.
C1	Original 8 uF; replacement must be a deliberate choice.

Notes

- Do not replace parts merely because they are old.
Replace if they measure wrong, leak, are mechanically faulty or visibly damaged.

Record

Point	Target	Measured	Comment
R35	10 k		
R19 total	2.96 Mohm		
R15	3.54 k		
RV5	10 k linear		
RV3	10 k linear		
C1	OK		

Step 3 - Mains setting and SET ~

Purpose

SET ~ is the base setting for the transformer primary/secondary levels. If SET ~ is wrong, later voltages and meter functions will be wrong.

Setup / settings

- No valve in socket.
- Correct mains tap/coarse mains selector chosen.
- CIRCUIT SELECTOR to CHECK (C).
- METER SWITCH to an anode-current range, not mA/V or D/R.
- LEAKAGE to ~.

Procedure

1. Switch on the instrument and allow it to stabilise briefly.
2. Turn the front-panel SET ~ control until the pointer is in the SET ~ zone/red line.
3. Use the front-panel SET ~ as the preliminary base setting before voltage checks.
4. After anode voltage has been verified as correct, adjust RV6 only if the meter pointer is not centred in the SET ~ zone. RV6 is the internal SET ~ calibration, not a general compensation for other faults.
5. Warning: on the MK III, RV6 is in a dangerous position in the bottom of the instrument. Do not perform live adjustment without a safe method, insulated trimming tool and full control over mains and HT voltages.

Target values / assessment

Point	Target value / comment
CIRCUIT SELECTOR	CHECK (C).
METER SWITCH	Anode-current range.
LEAKAGE	~.
Result	Pointer in SET ~ zone/red line.
RV6	Internal SET ~ calibration after anode voltage is verified. Not fault compensation.

Notes

- The R19 calibration resistor must not be altered just to make SET ~ look correct.

Record

Point	Target	Measured	Comment
SET ~	Red line/SET zone		
Mains tap	Closest correct		
RV6 adjusted?	No/Yes		

Step 4 - Pre-check of anode, screen, grid and heater

Purpose

Sanity-check that voltages exist, follow a reasonable sequence and show no obvious fault before fine calibration.

Setup / settings

- No valve in socket.
- SET ~ completed according to step 3.
- DMM in the correct AC/DC mode.
- Well-insulated probes, preferably crocodile clips.
- Know whether the instrument is original or has an anode-rectifier modification.

Procedure

1. Check heater voltages unloaded. They may be a few percent high with no load.
2. Set ANODE VOLTS to 400 V and put the instrument in TEST.
3. Measure between A1 and C. Original MK III is measured as AC RMS; do not expect 400 V DC.
4. In the original circuit, the 400 V range is about 444 V RMS unloaded.
5. Screen voltage is taken from the same HT taps as anode. Therefore check only presence, reasonable rising sequence and obvious faults; do not treat it as a separate precision calibration point in an original MK III.
6. Only check that grid voltage exists; exact grid calibration is done in step 6.
7. Switch off and record all values.

Target values / assessment

Point	Target value / comment
Heater unloaded	Typically slightly high with no load.
Anode 400 V original	approx. 444 V RMS between A1 and C.
Screen	Sanity-check: presence and reasonable sequence. Accurate screen accuracy normally does not need a separate check if anode voltage is correct.
Grid	Exact calibration in step 6.

Notes

- The two D77/6AL5 valves in the original MK III are used for grid and screen functions; they are not normal smoothed anode rectifiers for anode volts.
- If the instrument has been modified with additional diodes, interpret measured values according to that specific modification.

Record

Point	Target	Measured	Comment
Heater 6.3 V	Reasonable unloaded		
Anode 400 V	approx. 444 V RMS		
Screen	Sanity-check, not separate calibration point		
Grid sanity	Present		

Step 5 - Anode current with dummy load

Purpose

Check anode-current indication with a defined external DC current. This is a check measurement, not a normal internal adjustment.

Setup / settings

- Dummy load: 1 kV silicon diode + 3.9 k at least 10 W + 1 k approx. 3 W wirewound pot in series.
- Practical interpretation of diode direction: diode anode to A1; band/cathode towards resistor and C side. The reference text lists the dummy-load parts and A1-C connection, but does not explicitly describe the diode direction in words.
- External DC mA meter in series in A1 ANODE LINK. Open the link and replace it with the meter.
- METER SWITCH to 100 mA anode-current range.
- SCALE switch to OUTER.
- No valve in socket. Start with ANODE VOLTS at 0 V.

Procedure

1. Connect dummy load between A1 and C: A1 -> |>| -> 3.9 k/10 W -> 1 k pot -> C. The band/cathode should face the resistor/C side so the diode conducts when A1 is positive relative to C.
2. Open A1 ANODE LINK and place the external DC mA meter in series where the link normally sits.
3. Set METER SWITCH to 100 mA and SCALE to OUTER.
4. Put the VCM in TEST with ANODE VOLTS at 0 V.
5. Slowly increase ANODE VOLTS towards 400 V while watching the external anode current.
6. Adjust the dummy-load pot until the external meter reads exactly 50 mA DC.
7. The AVO meter should then indicate 100 mA +/- 5 % on the outer 0-100 scale.
8. Reduce ANODE VOLTS to 0 V, switch off and let the load cool.

Target values / assessment

Point	Target value / comment
External DC current	50 mA DC in A1 ANODE LINK.
AVO indication	100 mA +/- 5 %, i.e. about 95-105 mA.
Scale	OUTER scale, outer 0-100 mA scale.
Diode	1 kV Si diode. Practical interpretation: band/cathode towards C side. 1N4007 is suitable as a test diode.

Notes

- If step 5 is wrong: do not adjust RV4/RV6 or other trimmers to hide the fault. Troubleshoot SET ~, meter/R35, range resistors, switches, dummy load and external meter.
- The test load can get hot; keep the test brief.
- The diode direction above is a practical inference from the requirement for positive DC anode current from A1 through the load towards C. If no external DC current occurs with correct setup, switch off and first check diode direction and A1-link meter connection.

Record

Point	Target	Measured	Comment
External current	50.0 mA DC		
AVO indication	100 mA +/- 5 %		
SCALE	OUTER		
Diode direction	Practical interpretation: band towards C side		

Step 6 - Negative grid voltage

Purpose

Calibrate NEG. GRID VOLTS. The MK III uses coarse steps 0/20/40/60/80 plus variable RV5 scale 0-21. The calibration factor is 0.52.

Setup / settings

- No valve in socket.
- SET ~ and step 4 completed.
- Grid Link/Vg Link open.
- DMM on DC V, high input impedance, not LoZ.
- Measure between G and C. Black-lead to C and Red-lead to G gives negative reading for grid.
- Instrument in TEST mode so the grid supply is active.
- ANODE VOLTS, SCREEN VOLTS and HEATER preferably 0/off during this measurement.

Procedure

1. Switch off and open Grid Link/Vg Link.
2. Connect DMM between G and C.
3. Switch on, set SET ~ correctly and go to TEST.
4. Set incremental NEG. GRID VOLTS to 80 and variable NEG. GRID to 21 on the scale.
5. Target is -52.5 V between G and C.
6. If wrong: carefully adjust RV4 to -52.5 V using a safe method.
7. Set incremental to 0 and RV5 to 5. Check -2.6 V.
8. Set RV5 to 15. Check -7.8 to -7.9 V.
9. If the 0+5 point is wrong after 80+21 is correct, first check mechanical indexing/skirt of the NEG. GRID VOLTS knob. Do not chase the low scale with RV4 if the top point is already correct.
10. If the 0+15 point is too high despite correct 80+21 and correct indexing, the service procedure mentions correction with a fixed resistor between the zero end and slider of the grid pot. Do this only after R7-R15, RV5 and wiring have been verified.
11. Leave Grid Link open if step 7 is performed immediately; otherwise close it again.

Target values / assessment

Point	Target value / comment
80 + 21	-52.5 V (+52.5 V if meter leads are reversed)
0 + 5	-2.6 V.
0 + 15	-7.8 to -7.9 V.
20 + 6	approx. -13.5 V as practical check value.
DMM	At least approx. 2 Mohm; a normal 10 Mohm DMM is OK.

Notes

- 21 is the scale marking, not necessarily the mechanical end stop.
- If RV5/grid pot has been replaced, mechanical shaft position, skirt and stop position are as important as resistance value.

Record

Point	Target	Measured	Comment
Grid 80+21	-52.5 V		
Grid 0+5	-2.6 V		
Grid 0+15	-7.8 to -7.9 V		
Grid Link afterwards	Open for step 7 / otherwise closed		

Step 7 - mA/V calibration

Purpose

Check the mA/V function with small defined grid-voltage changes. The 220 k resistor is a temporary series resistor in the grid-pot feed, not a resistor between G and C.

Setup / settings

- Grid Link open from step 6.
- DMM between G and C on DC mV/V.
- Temporary 220 k in series with the live end of RV5/grid-pot feed must be fitted for this procedure. It must not be placed between G and C, and must be removed after step 7.
- SET mA/V control accessible.
- Instrument stable and SET ~ correct.
- Switch off and fit the 220 k series resistor in the live end of RV5/grid-pot feed before starting the mA/V check. Leave it fitted through all three mA/V points.

Procedure

1. With METER SWITCH on 2.5 mA, adjust NEG. GRID VOLTS until the DMM reads 104 mV.
2. Set METER SWITCH to mA/V and SET mA/V to 5 mA/V on the outer scale. DMM should be near 0 mV.
3. Return to 2.5 mA and set grid to 52 mV.
4. Set METER SWITCH to mA/V and SET mA/V to 10 mA/V on the outer scale. DMM near 0 mV.
5. Return to 2.5 mA and set grid to 0.52 V.
6. Set METER SWITCH to mA/V and SET mA/V to 1 mA/V on the outer scale. DMM near 0 mV.
7. Adjust the mA/V dial only if absolutely necessary and choose the best overall characteristic.
8. Radio Bygones gives no official tolerance for the zero reading; practically, the external voltmeter should be within a few millivolts of zero because mains fluctuations make better precision difficult.
9. Close Grid Link/Vg Link and remove the temporary 220 k resistor.

Target values / assessment

Point	Target value / comment
104 mV	SET mA/V 5; DMM near 0 mV.
52 mV	SET mA/V 10; DMM near 0 mV.
0.52 V	SET mA/V 1; DMM near 0 mV.
Important	52 mV, not 52.5 V.
220 k	Temporary series resistor, removed afterwards.

Notes

- If RV3 or RV5 has been replaced, repeat both grid and mA/V checks.

Record

Point	Target	Measured	Comment
mA/V 5	near 0 mV		
mA/V 10	near 0 mV		
mA/V 1	near 0 mV		
Grid Link	Closed afterwards		
220 k	Removed afterwards		

Step 8 - Resistance and leakage scale

Purpose

Check the insulation/leakage scale with an octal test plug and defined high-value resistors.

Setup / settings

- Octal plug with all resistors to common pin 1.
- 1 % resistors: 0.5 / 1 / 2.5 / 5 / 10 / 25 Mohm.
- Roller setting 145 678 900.
- Instrument correctly SET ~.

Octal socket electrode mapping:

Octal pin	Resistor to pin 1	Electrode / comment
1	Common / 0 ohm	C
2	0.5 Mohm	G
3	1 Mohm	S
4	2.5 Mohm	A1
5	5 Mohm	A2
6	10 Mohm	D1
7	25 Mohm	D2
8	NC	Not used for MK III leakage test

Procedure

1. Build the test plug: pin 1 common, other pins each with one resistor to pin 1.
2. Set roller setting 145 678 900.
3. Set the instrument according to the leakage/insulation procedure without a valve.
4. Go through the relevant LEAKAGE switch electrode positions.
5. Check that the scale gives the correct sequence and reasonable positions for the resistors.
6. For systematic error: check calibration resistors and meter circuit. For a single error: check switch contact, test plug and the current resistor.

Target values / assessment

Point	Target value / comment
0.5 Mohm	Low part of scale.
1 Mohm	Classic AVO check point.
2.5 Mohm	Practical mid-scale value.
5/10/25 Mohm	Upper-scale check.
Rollers	145 678 900.

Notes

- Keep fingers and contamination away from high-value nodes. Skin resistance and humidity affect Mohm values.

Record

Point	Target	Measured	Comment
0.5 M	0.5 Mohm		
1 M	1 Mohm		
2.5 M	2.5 Mohm		
5 M	5 Mohm		
10 M	10 Mohm		
25 M	25 Mohm		

Step 9 - Cut-out relay

Purpose

Check that the cut-out relay trips under controlled overcurrent/short-circuit conditions.

Setup / settings

- Instrument stable after previous steps.
- No valve in socket.
- Short insulated test lead or controlled switch/jig.
- Be ready to interrupt immediately.

Procedure

1. Set ANODE VOLTS to 100 V.
2. Activate test mode according to normal procedure.
3. Briefly and controllably short A1 to C. The relay should trip.
4. Reset relay/instrument.
5. Repeat with A2 to C.
6. Set SCREEN VOLTS to 60 V.
7. Briefly and controllably short S to C. The relay should trip.
8. If the relay does not trip: troubleshoot relay circuit and contact function before using the instrument with valves.

Target values / assessment

Point	Target value / comment
A1-C at ANODE 100 V	Relay trips.
A2-C at ANODE 100 V	Relay trips.
S-C at SCREEN 60 V	Relay trips.
Adjustment	Normally none if the relay works.

Notes

- This is a function test, not a stress test. Shorts must be brief and controlled.

Record

Point	Target	Measured	Comment
A1-C	Trips		
A2-C	Trips		
S-C	Trips		

Step 10 - D/R test

Purpose

Check diode/rectifier function with a simple series jig. Keep the test short to avoid overload.

Setup / settings

- 150 ohm at least 5 W in series with silicon diode.
- 1N4007 or equivalent 1 kV diode.
- Short, insulated test jig.
- AVO correctly SET ~.

Procedure

1. Build a series connection of 150 ohm/5 W and diode.
2. Connect the test jig between D1 and C according to the D/R procedure.
3. Set METER SWITCH to the relevant D/R range, e.g. D/R 180 or D/R 60 depending on the check.
4. Briefly perform the test on D/R 180. The cut-out relay must not trip.
5. Move the test jig to D2 and C and repeat on D/R 180. Then repeat both D1-C and D2-C on D/R 60.
6. If the relay trips or the indication is unreasonable: check diode polarity, test jig, switch settings, ELECTRODE SELECTOR position and D/R circuit before using the instrument further.

Target values / assessment

Point	Target value / comment
Series resistor	150 ohm, at least 5 W.
Diode	Silicon diode, correct polarity.
Test points	D1-C and D2-C.
Test duration	Very brief. The diode test must not be prolonged because R37/load resistor can be damaged.

Notes

- AVO/Radio Bygones warn that the diode test must not be prolonged; otherwise R37/load resistor may be overloaded. This does not replace complete testing of a real rectifier valve under all operating conditions.

Record

Point	Target	Measured	Comment
D1-C D/R 180	Relay must not trip		
D2-C D/R 180	Relay must not trip		
D1-C D/R 60	Relay must not trip		
D2-C D/R 60	Relay must not trip		

Step 11 - Final check with known valve

Purpose

Verify the whole chain with a known good reference valve after internal check points are complete.

Setup / settings

- Known good, common reference valve.
- Historical AVO reference: strapped CV491/ECC82/12AU7 at 16 mA anode current was used as a standardised check valve. Use this as a reference idea, not as a requirement if you do not have a truly standardised valve.
- Correct row in the AVO Valve Data Manual or your Excel table, checked against the original PDF.
- Correct socket and possible top cap.
- Instrument warmed up and SET ~ adjusted.

Procedure

1. Choose a valve with known history, not an expensive collector valve as the first test.
2. Look up selector switch, top cap, Vh, Vg1, Va, Vg2, Ia and gm.
3. Check socket and pin numbering; many socket diagrams are shown from below.
4. Set rollers and controls with the instrument in a safe idle state.
5. Perform the normal AVO sequence: SET ~, leakage/shorts, heater continuity and then test.
6. Check backing-off, Ia and mA/V/goodness.
7. Compare with previous known values or reasonable valve data. AVO results do not need to match a modern DC curve tracer exactly.

Target values / assessment

Point	Target value / comment
Valve data	Checked against VDM/original source. AVO mentions strapped CV491/ECC82/12AU7 at 16 mA as historical reference.
SET ~	Set before test.
Leakage/shorts	OK before Ia/mA/V.
Ia and mA/V	Reasonable against valve data and reference.

Notes

- If a known valve deviates strongly: go back to previous steps and look systematically rather than trimming by feel.
- Even standardised valves can deviate more than the service manual nominal tolerances; use the final check to detect systematic faults, not to force the instrument to one single valve.

Record

Point	Target	Measured	Comment
Valve type			
Ia	Reasonable vs data		
mA/V	Reasonable vs data		
Leakage	OK		

Step 12 - Calibration record and close-out

Purpose

Collect final values, instrument data and replaced components so the instrument is easier to troubleshoot later.

Setup / settings

- All previous steps performed or marked not performed.
- Instrument serial number.
- Measurement equipment model and calibration status if known.
- List of replaced components and old measured values.

Procedure

1. Record instrument serial number, date and measurement equipment.
2. Record mains voltage and SET ~ position during calibration.
3. Collect results from steps 1-11.
4. Mark all replaced components and keep old measured values if possible.
5. Record especially meter protection, C1 replacement or rectifier changes.
6. Perform visual check: no loose wires, no tools left inside, correct insulation and no exposed conductors near chassis.
7. Refit panels and covers.
8. Perform final SET ~ and brief sanity-check with reference valve after reassembly.

Target values / assessment

Point	Target value / comment
Meter FSD	30 μ A.
Meter resistance	3250 ohm +/- 1 %.
SET ~	Red line/SET zone.
Anode 400 V	approx. 444 V RMS original.
Anode dummy	50 mA external -> 100 mA on AVO.
Grid 80+21	-52.5 V.
mA/V	DMM near 0 mV at three points.
Relay	Trips in cut-out test; does not trip in D/R 180/60 with 150 ohm + diode.

Notes

- Store this document together with the instrument schematic, layout, photos and component list.

Record

Point	Target	Measured	Comment
Date			
Serial number			
Measurement equipment			
Replaced components			
Final status	Pass / fail		

5. References

- Euan MacKenzie, "Servicing and Calibrating AVO Valve Characteristic Meters MK III and MK IV", Radio Bygones No. 153, February/March 2015.
- Martin Forsberg & Euan MacKenzie, "AVO VCMs - circuit comparisons", 2012-07-31.
- AVO Valve Data Manual, edition 23, Megger/AVO, 1981
- AVO Valve Characteristic Meter MK III Working Instructions / original AVO documentation.
- VCM MK III circuit diagram, corrected by Martin Forsberg, scan/version 2019.
- AVO VCM MK III layout PDF, plates for front panel, power supply panel, control panel, valve holder panel and movement assembly.

Appendix A - Circuit diagram

VCM MK III circuit diagram, corrected/2019 scan. Use as an overall reference for function and component designations.

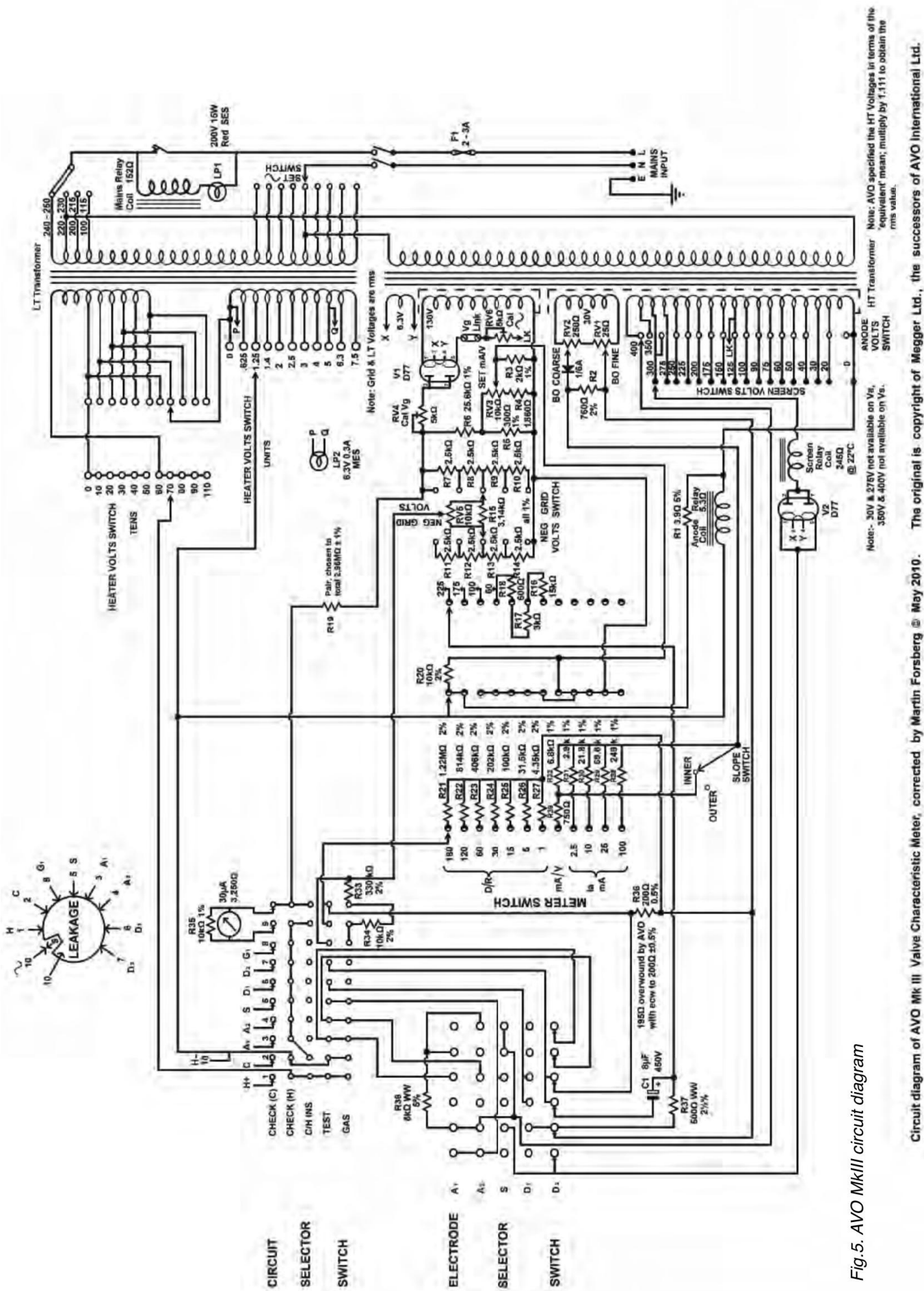


Fig.5. AVO MkIII circuit diagram

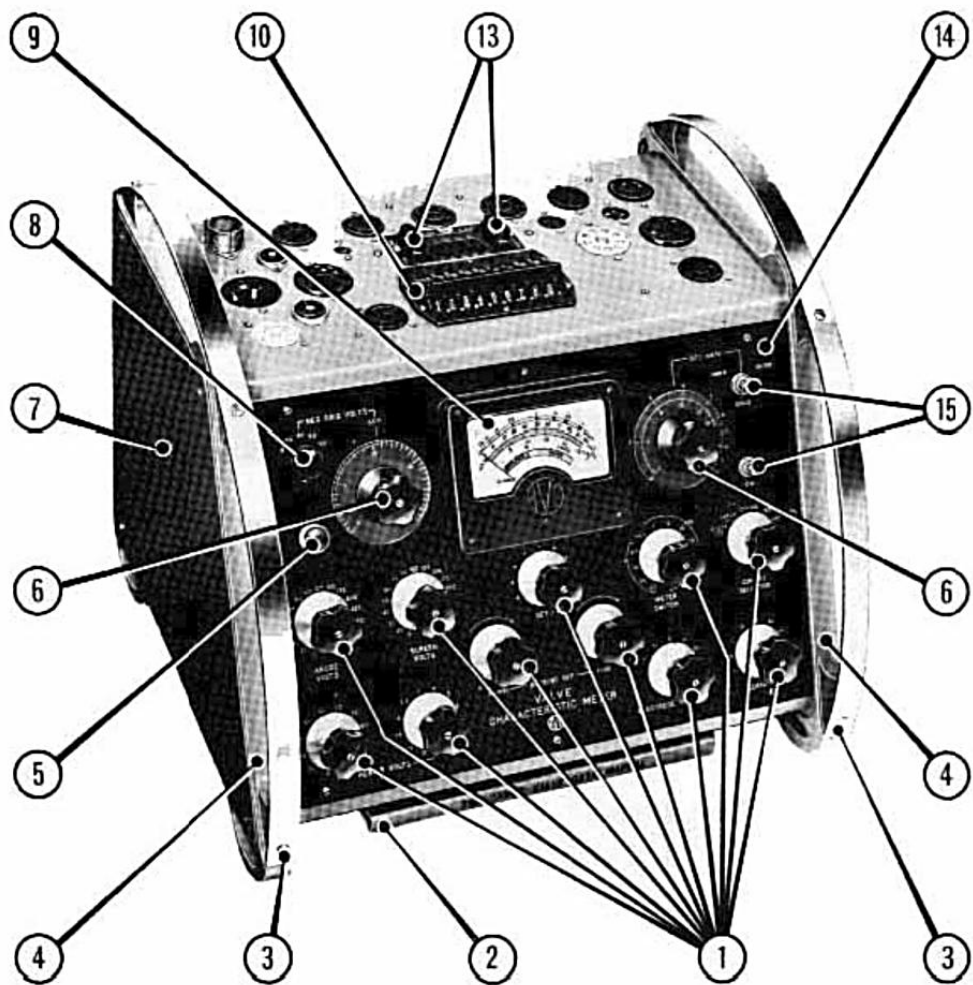
Appendix B - AVO VCM MK III layout

The layout shows panels, component locations and plate references.

Front Panel - Parts list

Item No.	Description	Circuit Ref	Part No.
1	Knob Assembly Complete with Skirt		15220-A
2	Valve Data Manual		-----
3	Handle		40533-2
4	End Frame		40530-3
5	Indicating lamp holder		14812-1
	Bulb 6.5V 0.3A	LP2	50010-14
6	Knob Assembly Complete with Dial		15220-C
7	End Plate		40529-3
8	Knob (Screw AS23)		12840-A
9	Meter Movement Complete		40650-C
10	Roller Selector Switch Escutcheon		40166-2
11	Top Cap Lead		11237-C
12	Valve Board Lid Assembly		40137-B
13	Anode Link Terminals		13834-A
14	Front Panel		40525-1
15	Toggle Switch		13657-1
*16	Knob For Item 1		14267-1
*17	Transparent Cursor For Item 6		14806-1

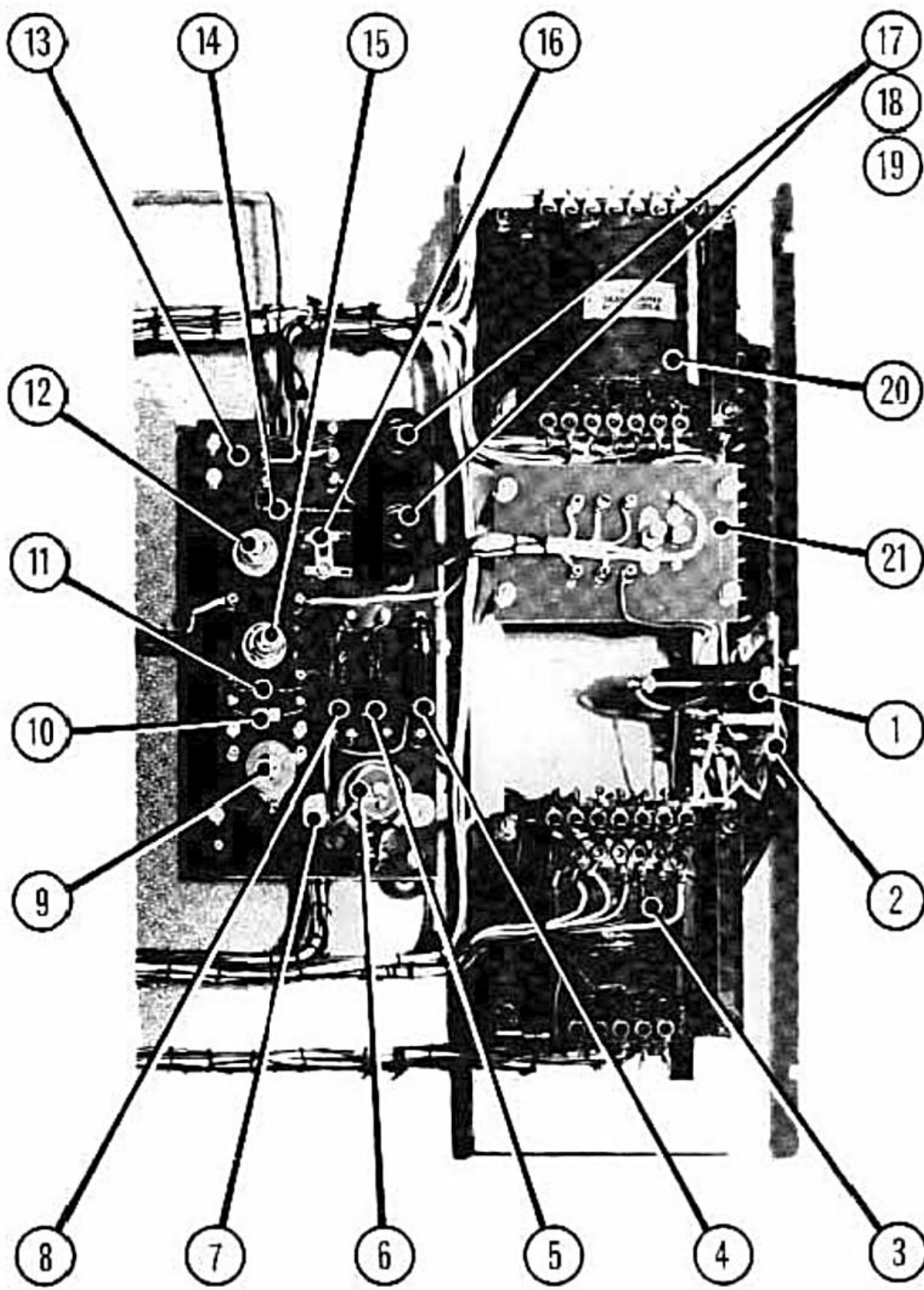
* Not illustrated.



Power Supply Panel - Parts list

Item No.	Description	Circuit Ref	Part No.
1	Fuse	F1	12239-3
2	Mains Selector Panel Assembly		11966-C
3	LT Transformer Assembly	LT1	21005-B
4	Resistor 200 Ω	R36	14709-B
5	Resistor 8k Ω $\pm 5\%$ Vitreous	R38	12049-400
6	Capacitor 8uF Electrolytic	C1	12049-404
7	Capacitor Clip		13699-1
	Capacitor Clip Rubber Strip		14772-1
8	Resistor 500 Ω $\pm 2\%$ Vitreous	R37	12049-401
9	Rectifier 1/6 A	REC 1.	12049-704
10	Resistor 3.9 Ω $\pm 20\%$	R1	12049-434
11	Resistor 760 Ω $\pm 2\%$	R2	12049-422
12	Potentiometer 5k Ω Set Grid Volts	RV4	10770-5
13	Tag Board (Tagged)		21004-25
14	Resistor 25.6k Ω $\pm 1\%$	R6	12049-420
15	Potentiometer 5k Ω Cal. Set ~	RV6	10770-5
16	Link		13062-2
	Self Tapping Screw		PKS4
17	Valve Holder with Skirt B7G		13738-2
18	Valve Holder Can B7G		13819-1
19	Valve Type D77	V1/V2	12241-S
20	HT & Grid Transformer Assembly	HT1	21002-B
21	Relay Assembly Complete		40466-A

Power Supply - Image



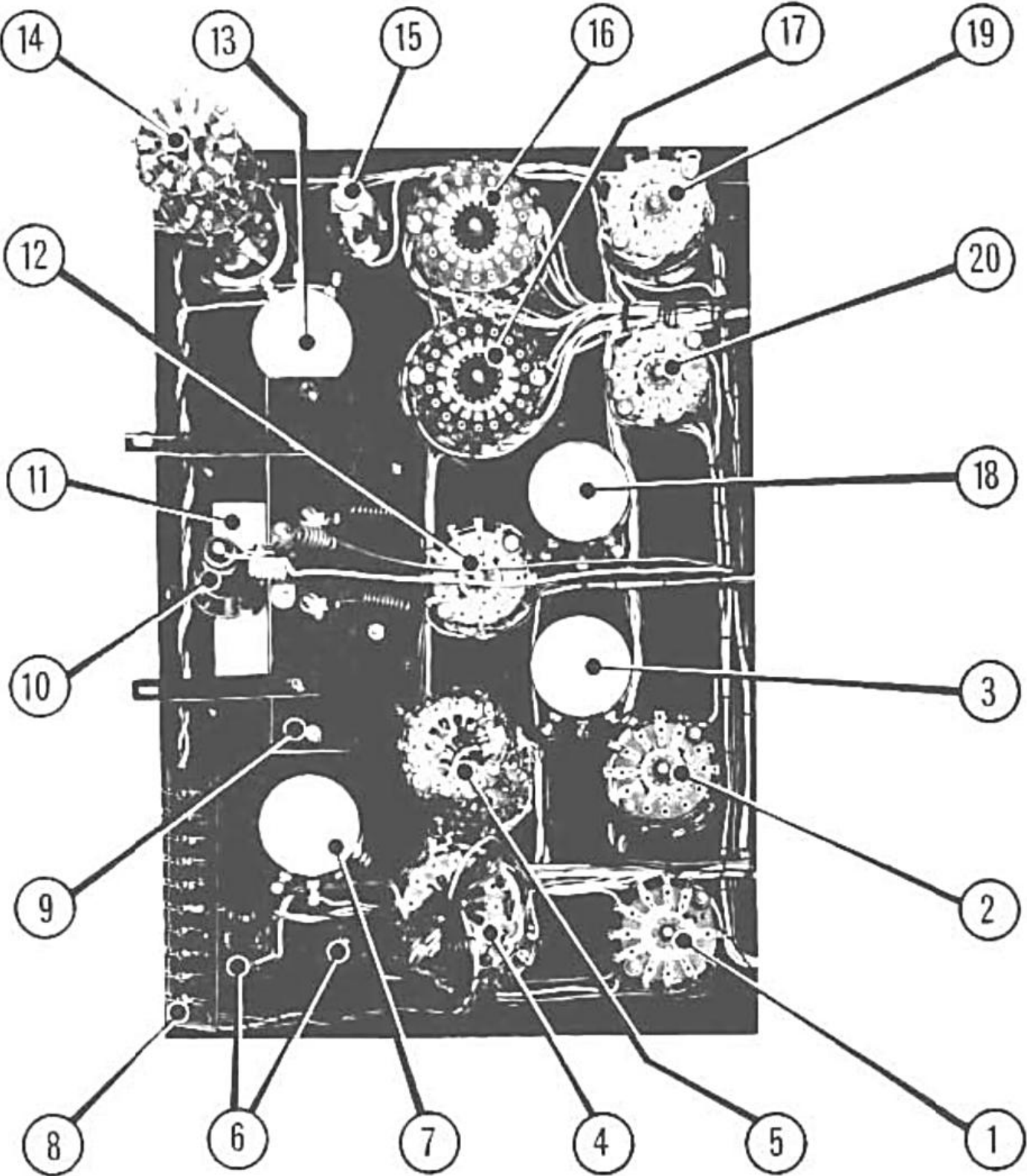
Control Panel - Parts list

Item No.	Description	Circuit Ref	Part No.
1	Leakage Switch		20992-1
2	Electrode Selector Switch		20996-1
3	Fine Backing Off Control 250 Ω	RV1	14558-4
4	Circuit Selector Switch		20991-1
	Resistor 15k $\Omega \pm 2\%$	R16	12049-391
	Resistor 2.96M $\Omega \pm 1\%$ (Matched Pair)	R19	12049-433
	Resistor 10k $\Omega \pm 1\%$	R35	12049-386
	Resistor 10k $\Omega \pm 2\%$	R34/R20	12049-385
	Resistor 330k $\Omega \pm 2\%$	R33	12049-384
	Resistor 2k $\Omega \pm 1\%$	R3	12049-424
	Resistor 1860 $\Omega \pm 1\%$	R4	12049-423
	Resistor 300 $\Omega \pm 1\%$	R5	12049-421
	Resistor 600 $\Omega \pm 2\%$	R18	12049-389
	Resistor 3k $\Omega \pm 2\%$	R17	12049-390
5	Meter Switch		20997-1
	Resistor 6.8k $\Omega \pm 1\%$	R32	12049-428
	Resistor 2.9k $\Omega \pm 1\%$	R31	12049-426
	Resistor 21.8k $\Omega \pm 1\%$	R30	12049-429
	Resistor 59.6 $\Omega \pm 1\%$	R29	12049-430
	Resistor 249k $\Omega \pm 1\%$	R28	12049-431
	Resistor 1.22M $\Omega \pm 2\%$	R21	12049-432
	Resistor 814k $\Omega \pm 2\%$	R22	12049-392
	Resistor 406k $\Omega \pm 2\%$	R23	12049-393
	Resistor 202k $\Omega \pm 2\%$	R24	12049-394
	Resistor 100k $\Omega \pm 2\%$	R25	12049-395
	Resistor 315k $\Omega \pm 2\%$	R26	12049-396
	Resistor 4.35k $\Omega \pm 2\%$	R27	12049-397
6	Toggle Switch		13657-1
7	Potentiometer 10k Ω lin, "Set mA/V"	RV3	14558-3
8	Tag Board Assembly		11996-A
9	Movement Board Assembly		14810-2
10	Lamp	LP1	14653-1
11	Meter Movement Complete		40650-C
12	Set ~ Switch		20994-1
13	Potentiometer 10k Ω lin, "Neg. Grid Volts"	RV5	14558-3
14	Negative Grid Volts Switch		20995-1
	Resistor 2.5k $\Omega \pm 1\%$	R7/R14	12049-425
	Resistor 3.54k $\Omega \pm 1\%$	R15	12049-427
15	Indicating lamp holder		14812-1
	Bulb 6.5V 0.3A	LP2	50010-14
16	Anode Volts Selector Switch		14822-1
17	Screen Volts Selector Switch		14822-1
18	Coarse Backing Off Control 250 Ω lin	RV2	14558-5
19	Heater Volts Switch Coarse		20993-1
20	Heater Volts Switch Fine		20994-1

Note on RV3/RV5

In the original Control Panel parts list, the Circuit Ref column identifies Set mA/V as RV5 and Negative Grid Volts as RV3. This appears to be reversed compared with the circuit diagram and calibration procedure. In this revised table, the circuit references have therefore been corrected to match the schematic/function rather than the original parts -list entry.

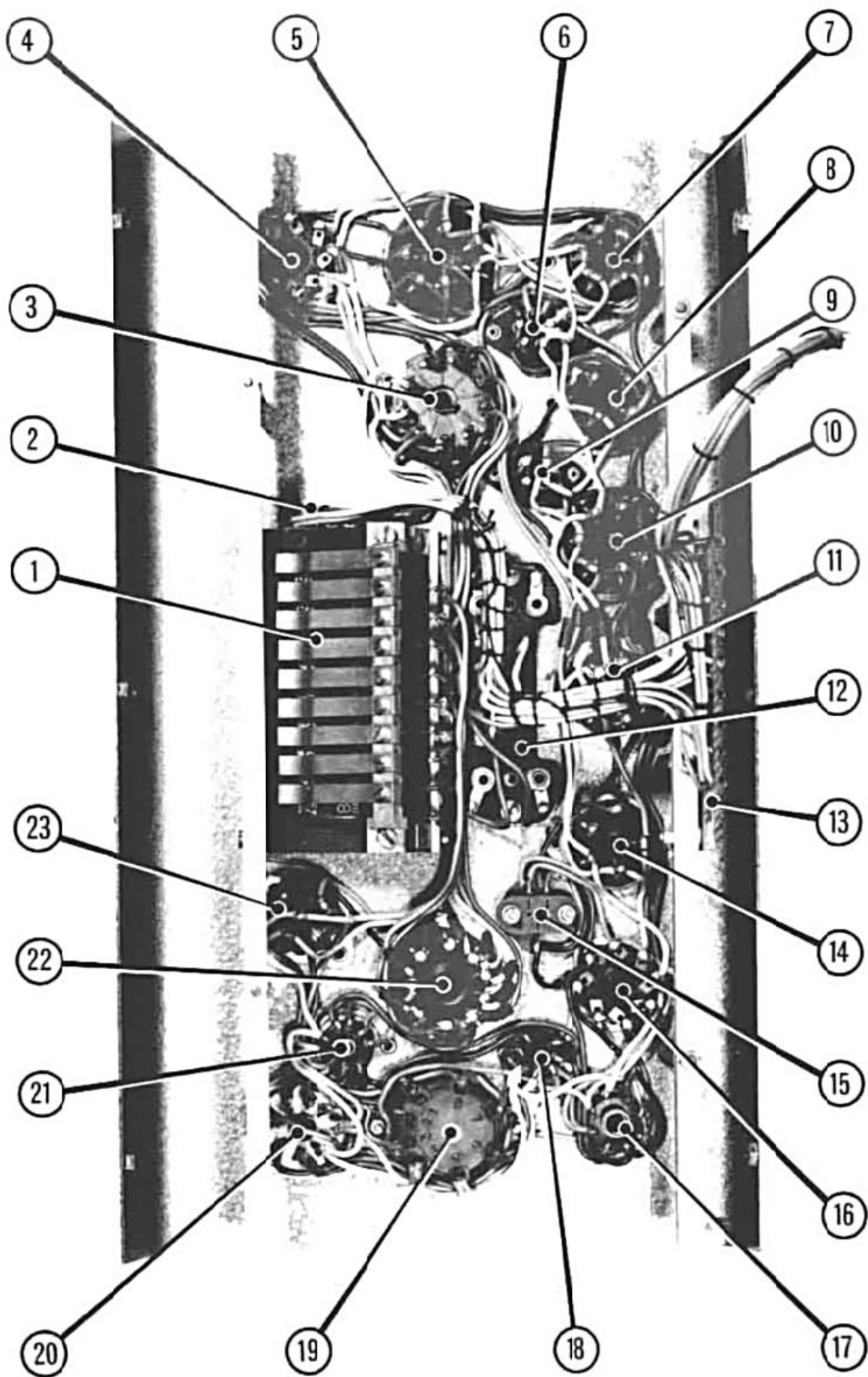
Control Panel - Image



Valve Holder Panel - Parts list

Item No.	Description	Circuit Ref	Part No.
1	Roller Selector Switch Assembly		40157-B
2	Ferroxcube Bead		14747-1
3	Valve Holder B9G		10281-1
4	Valve Holder IO8 International Octal		40140-8
5	Valve Holder B7 English 7 Pin		40140-2
6	Valve Holder SM5 Hivac 5 Pin		40140-13
7	Valve Holder UX6 American 6 Pin		40140-9
8	Valve Holder UX4 American 4 Pin		40140-4
9	Valve Holder SM4 Hivac 4 Pin		40140-12
10	Valve Holder UX7 American 7 Pin (Large)		49140-3
11	Valve Holder SM7 American 7 Pin (Small)		40140-20
12	Top Cap Escutcheon Assembly		20903-A
13	Tag Board Assembly		11996-A
14	Valve Holder UX5 American 5 Pin		40140-6
15	Valve Holder B3G		10509-1
16	Valve Holder MO8 Mazda Octal		40140-7
17	Valve Holder B8A		40140-11
18	Valve Holder B7G		40140-14
19	Valve Holder 8SC (P Type Base		40140-10
20	Valve Holder B8G		40140-16
21	Valve Holder B9A		40140-19
22	Valve Holder B9 English 9 Pin		40140-1
23	Valve Holder B4/5 English 4/5 Pin		40140-5

Valve Holder Panel - Image



Movement Assembly - Parts list & Drawing

Item No.	Description	Part No.
1	Scale plate	(14824-4)
		(14824-10)
2	Moving coil assembly	21124-D
3	Zero Adjustor	15436-1
4	Hairsprings	10075-16
5	Sprung Jewel Assembly	10184-8
6	Pivots	10158-4
7	Movement Case Front	40537-2
8	Movement Case Rear	40538-A
*9	Rear Case Glass	10059-B

* Not illustrated.

